

Understanding changes over time in the receipt of workers' compensation benefits

Marc Fulmer*
Michigan State University

February 12, 2026
[Click here for the most recent version](#)

Abstract

Workers' compensation, the oldest social insurance program in the United States, continues to serve as an important safety net for the vast majority of the workforce. The program supports workers at a particularly vulnerable time: immediately after suffering an occupational injury or illness. Yet, despite this vital role in supporting workers, less is known about the program's long-term trends in reciprocity. To understand these trends, I used a descriptive decomposition framework to systematically investigate the reasons for the 43.6 percent decrease in the reciprocity rate of cash benefits over the period from 2003 through 2018. I find that 68 percent of this decrease—29.5 percentage points—is associated with improvements to workplace safety. (JEL No. I18, J28, J32)

*Job Market Paper. Email: fulmerm1@msu.edu. Work on this paper was supported, in part, by the Early Start Program of the College of Social Science at Michigan State University.

1 Introduction

Workers' compensation (WC) is the oldest social insurance program in the United States (Fishback and Kantor 2000). Before the introduction of WC, workers suffering occupational injuries and illnesses (OIs) could seek remuneration only by filing lawsuits against their employers. However, workers frequently recovered nothing for their medical bills and lost wages due to employer legal defenses. For their part, employers faced the risk of extraordinarily large jury verdicts in the cases that they lost. This litigation-based system eventually proved untenable and led to a compromise in the early 20th century known as the "Grand Bargain." The bargain meant that workers gave up their ability to sue in exchange for employers providing medical and cash benefits to workers suffering OIs, regardless of fault.

This no-fault framework forms the basis of WC, which continues to offer widespread and fundamental protection for workers. In 2022, 93 percent of all employees, representing \$10.3 trillion in wages, were covered under WC (Welch et al. 2024). This breadth of coverage, combined with the assurance of payment for OIs, makes WC a foundational part of the labor market.

Because of its role in the labor market, WC has been studied extensively, yet less is known about the long-term trend in the rate of benefit receipt from the program. In large part, this stems from the scarcity of data on this topic at the national level, as WC is primarily a state-level program without a centralized clearinghouse. Despite this challenge, understanding these long-term trends is important because it would help stakeholders make more informed decisions on potential future changes to the program in their states.

Specifically, policy decisions should depend on identifying the mechanisms most associated with changes in reciprocity. Different underlying mechanisms can have very different implications for workers and for how the WC system is evaluated. For instance, a decrease in benefit receipt may be associated primarily with improvements to workplace safety such as reductions in serious injuries or effective adoption of preventive measures. In this case, maintaining current policy may be wise. Conversely, if a decrease in benefit receipt is primarily associated with increasingly stringent insurer decisions, stakeholders might need to reassess the amount of compensation paid

to workers with OIIs, the freedom with which workers have to choose their medical providers, or the burden of proof that workers need to meet to receive benefits. These interpretations differ sharply, so identifying the correct mechanism is essential for effective policymaking. Competing considerations such as these led to the question I answered in this paper: How has the rate of WC receipt changed over time and why has it done so?

I first analyzed data from the National Council on Compensation Insurance (NCCI), an organization that recommends WC insurance rates in 35 states plus the District of Columbia. Analysis of these high-quality data allowed for an understanding of the overall trends of the two broad types of benefits received under WC programs: payments for medical expenses only (“medical-only benefits”) and payments for both medical expenses and lost wages (“indemnity benefits”). Over the period from 2003 through 2018, I found that the rate of medical-only benefit receipt decreased by 38.5 percent. Similarly, I found that the rate of indemnity benefit receipt decreased by 33.1 percent.

To gain a deeper understanding of the mechanisms driving this decrease, I analyzed data from the Annual Social and Economic Supplement of the Current Population Survey (ASEC). While the ASEC is a household survey and prone to under-reporting (*e.g.*, due to *recall bias* or respondent confusion), it offers a panel component that allows for linking respondents across two consecutive years. By exploiting this design, I estimated a 43.6 percent decrease in the rate of indemnity benefit receipt from 2003 through 2018.

Thus, the decrease as estimated from the ASEC was consistent with the calculations based on data from the NCCI. Because of this, I proceeded with a systematic investigation of the decrease through decomposition analyses. The analyses separated the decrease in reciprocity into four sources: continued receipt, sectoral composition, OIIs, and other factors. It is important to note that the decompositions presented here were based on accounting identities, not causal estimates. It was unlikely that causal estimates could be obtained under any empirical approach given the setting of this paper, and therefore the approach I used provided the most rigorous method to determine the relative importance of the factors driving the decrease.

First, I decomposed the decrease into new receipt (workers first receiving WC benefits in their

second year of the ASEC panel) and continued receipt (workers receiving WC benefits in both years of their responding). I found that 87.9 percent of the decrease in total receipt was associated with decreases in new receipt and that 12.1 percent was associated with decreases in continued receipt. Decreases in each type of receipt may have occurred for a variety of reasons. The decrease in new receipt may have occurred due to the changing composition of the economy from hazardous goods-producing industry sectors, such as Manufacturing, to safer service-providing industry sectors, such as Education and Health. Alternatively, the decrease may have occurred due to a reduction of the rates of OIIs, whether through genuine safety improvements or employer strategies to minimize recordable incidents. On the other hand, the decrease in continued receipt could have occurred due to OIIs becoming less severe over time or a decrease in the duration of benefits allowed by a WC program.

Next, I examined the explanation above that the decrease in new receipt was associated with the changing composition of the economy. To do so, I decomposed changes in new receipt into those resulting from changing sectoral composition versus changes occurring within industry sectors. I determined that 95.9 percent of the decrease in new receipt (84.3 percent of the decrease in total receipt) occurred due to within-sector effects. This finding suggests that the decrease is not meaningfully associated with the structural transformation of the economy.

Finally, I investigated if the decrease in new receipt was correlated with improvements to workplace safety. Using data from the Survey of Occupational Injuries and Illnesses (SOII), I concluded that 67.6 percent of the decrease in total receipt was associated with decreases in the rates of OIIs. This indicated that the most important contributor to the 43.6 percent decrease in total receipt was the broad-based decrease in OIIs. This association could have occurred due to genuine improvements to workplace safety from sources such as increased use of automation and new regulations from the Occupational Safety and Health Administration. Moreover, the association could reflect some degree of employers becoming more aggressive at returning employees to work or minimizing recordable OIIs.

My paper contributes to the literature by providing the first systematic exploration of the reasons

behind the long-term trends in WC receipt. In doing so, my paper complements prior work that has examined WC over the business cycle ([Hartwig et al. 1997](#); [Dong and Jestrab 2022](#)) and research observing that OIIs respond procyclically to economic conditions ([Asfaw et al. 2011](#); [Charles et al. 2022](#)). I extend this literature by showing the long-term drivers of reciprocity that persist beyond cyclical fluctuations. Additionally, I extend the results of previous studies that found that changes to sectoral composition do not serve as a significant explanation for decreases in the rate of OIIs ([Ruser 2014](#); [Cohn and Wardlaw 2024](#)). Lastly, my results show a potentially substantial role that behavioral factors, such as worker responses to changing insurer behavior ([Biddle 2001](#)) and potential employer retaliation ([Biddle and Roberts 2003](#)), play in influencing reciprocity. These remain an important avenue of inquiry for explaining why rates of reciprocity have decreased so markedly over time.

This paper proceeds as follows. Section 2 provides the background of the WC program. Section 3 describes the datasets that I use in the paper. Section 4 answers *how* WC receipt changes over time. Section 5 answers *why* WC receipt changes over time. Section 6 concludes.

2 Background

WC is primarily a state-based system.^{1,2} The state-by-state implementation of WC programs began in Wisconsin in 1911 and continued until Mississippi implemented its program in 1948. Every state in the nation (and Washington, D.C.) has continued to administer its own WC program since its initial implementation. The key aspect of WC is that it is a no-fault system of insurance that pays regardless of whether an employer or employee is responsible for an OII. Forty-seven states and D.C. require employers to provide WC insurance coverage to their employees, with limited exceptions. On the other hand, South Dakota, Texas, and Wyoming do not mandate that employers provide this coverage. The upshot for workers covered under WC policies is that it serves as the

¹The discussion in this section is based on the report by [Welch et al. \(2024\)](#).

²Federal programs cover federal employees, nuclear weapons workers diagnosed with certain conditions due to radiation, beryllium, or silica exposure, coal mine workers diagnosed with Black Lung Disease, land-based maritime workers, and railroad workers.

“exclusive remedy” for OIIs. This means WC precludes the filing of lawsuits seeking additional remuneration except for gross negligence by an employer.

Employers may obtain WC coverage in three main ways, although the available options depend on the law in a given state. First, all states except North Dakota, Ohio, Washington, and Wyoming permit employers to purchase coverage through private insurers. Second, all states except North Dakota and Wyoming permit employers with adequate resources to self-insure their WC risk. Third, 21 states offer coverage through state-administered insurance funds.³ In two of these 21 states—North Dakota and Wyoming—employers must purchase their coverage through the state-administered fund. In another two of these 21 states—Ohio and Washington—employers must either purchase their coverage through the state-administered fund or self-insure.

Broadly, the program provides two types of benefits: payment for all OII-related medical expenses and payment for part of a worker’s lost wages due to an OII. These latter type of benefits are referred to as *indemnity* benefits. In general, workers who receive indemnity benefits also receive payment of their OII-related medical expenses. However, the reverse is not true. Indeed, approximately 75 percent of all recipients of WC benefits receive “medical-only” payments.

The amount of indemnity payments an injured worker receives depends on OII duration and extent along with state-specific parameters. With respect to duration, there are two categories of benefits: temporary and permanent. Temporary benefits are paid during the initial days after an OII, after a waiting period of between three to seven days, depending on the state. At the end of a worker’s recovery from their OII, they reach so-called *maximum medical improvement*. Impairments remaining after this point may qualify for permanent benefits. With respect to extent, there are two categories of benefits: partial and total. Partial benefits are paid when workers are able to work, but with a loss in work capacity. Total benefits are paid when workers are completely unable to work. Since indemnity benefit types combine both duration and extent, there are benefits for temporary partial, temporary total, permanent partial, and permanent total disabilities. Additionally, WC pays

³These states are California, Colorado, Hawaii, Idaho, Kentucky, Louisiana, Maryland, Missouri, Montana, New Mexico, New York, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Texas, Washington, and Wyoming.

cash benefits to a workers' family and dependents for fatal OIIs.

States differ widely in their compensation parameters, which are defined by the legislature in each state. In January 2021, for instance, Arkansas had a weekly benefit range of \$20 to \$736 while Iowa had a weekly benefit range of \$326 to \$1,864 for temporary total disabilities. States often base compensation for temporary partial disability on two-thirds of a worker's pre-OII weekly wage, but Michigan bases its calculation on 80% of spendable earnings. Indiana caps PPD benefits for many OIIs at \$390,000, yet Oregon has no such cap. Finally, Alaska limits permanent total disability to 300 weeks, but 32 states have no limit at all.

3 Data

3.1 National Council on Compensation Insurance

The National Council on Compensation Insurance (NCCI) is a nonprofit organization that standardizes the calculation of WC insurance rates across 35 states and the District of Columbia (each of these jurisdictions is colloquially referred to as an "NCCI state"). In addition to these NCCI states, two states, Indiana and North Carolina, report data to the NCCI, bringing to a total of 38 jurisdictions that report data to NCCI. One key output of NCCI consists of state-specific insurance rate recommendations for approximately 550 classification codes to which employers are assigned based on their business operations. These insurance rates then become the basis of the experience-rated premiums that employers pay to obtain WC insurance coverage for their employees.

As part of its work, NCCI receives data from insurers. Insurers providing WC coverage in NCCI states are required by their state to report this data to NCCI. The data insurers report includes, among other items, data on the medical and indemnity benefits that workers receive. Based on this data, NCCI separately publishes an *Annual Statistical Bulletin*. One exhibit in this report lists the amount, per 100,000 insured workers, of new recipients of medical-only or indemnity benefits for all 38 jurisdictions that report data to NCCI. I call the values in this table "NCCI Data" in the remainder

of this paper. While the NCCI Data do not cover every state, these data are of high quality and allow for analysis of the two key types of WC benefits.

3.2 Annual Social and Economic Supplement

The Current Population Survey (CPS) is conducted by the Census Bureau on behalf of the Bureau of Labor Statistics (BLS). In the United States, it forms the basis of estimating the unemployment rate. To conduct the CPS, the Census Bureau administers this *voluntary* survey to a nationally representative sample of over 60,000 households. A given household is surveyed over 16 months in a 4-8-4 rotating panel: in the CPS for four months, out of the CPS for eight months, and back in the CPS for the final four months. Along with labor force status (employed vs. unemployed), the survey captures demographic information such as age, race, and education.

In addition to the basic CPS described above, households fill out supplemental surveys to provide richer information. The most well-known and widely used of these supplements is the Annual Social and Economic Supplement (ASEC). The Census Bureau administers the supplement each year in February through April (it is also known informally as the “March CPS”). Households that provide information on the ASEC include all of the households being sampled for the basic CPS in March. To improve estimates for certain demographic groups along with estimates on coverage under the Children’s Health Insurance Program, the Census Bureau surveys additional households, as well. Importantly, individuals can respond to the ASEC in two consecutive years.

Using the ASEC, a researcher can obtain data on poverty, geographic mobility, health insurance, and income from a variety of sources, including WC. Previous studies that have investigated various aspects of WC using the ASEC (e.g., benefit generosity, [Bronchetti and McInerney 2012](#); interaction with medical marijuana legalization, [Ghimire and Maclean 2020](#); interaction with recreational marijuana legalization, [Abouk et al. 2023](#); interaction with paid sick leave programs, [Dong et al. \(2024\)](#)) support the use of the ASEC for WC analyses. While the ASEC covers only indemnity benefits, the data represents all states, and, importantly, allows for an investigation at the industry sector level.

Since I needed to separate within-sector changes in WC receipt from changes occurring due to changing sectoral composition, having a consistent coding of industry sector across time was essential. I used a variable that IPUMS CPS ([Flood et al. 2024](#)) created for this purpose and developed a crosswalk between it and the industries as defined by the BLS for its programs, including the Survey of Occupational Injuries and Illnesses. The crosswalk makes it straightforward to aggregate and analyze the ASEC data by sector.

The number of employees by sector by state by year represent explanatory variables in the next section. While the National Academy of Social Insurance produces estimates of employment covered by WC in each state, these are not disaggregated at any level ([Welch et al. 2024](#)). Hence, my best estimate of WC coverage by sector would be the number of employees by state by year. This is because, except for the limited exceptions described in Section 2, employees are required to be covered by WC in almost every state. Indeed, state-year estimates of the total number of employees from the ASEC have a correlation with the estimates from [Welch et al. \(2024\)](#) of over 0.99.

In the ASEC, not every respondent in a given year 1 is linked to a response in year 2. Because of this, I constructed two samples from the ASEC to study receipt of WC over time in this section. The first sample I called the Full Sample. The Full Sample included all respondents each year, including those that were not able to be linked. The second sample I called the Linked Sample. The Linked Sample included only those respondents who were successfully linked between their two years of responding to the ASEC. Additionally, the Linked Sample considered only the second year that these successfully-linked respondents responded to the ASEC.

There are two important considerations when using the Linked Sample to compare to either the Full Sample or the NCCI Data. These considerations arise because the Linked Sample permits analysis of two different types of receipt: new receipt (respondent not a recipient in the first year of response, but is a recipient in the second year) and continued receipt (respondent a recipient in both years). The first consideration is that I cannot distinguish between the types of receipt in the Full Sample for all respondents. Hence, when I compared the Linked Sample to the Full Sample,

I considered both new and continued receipt. The second consideration is that the NCCI Data is most similar to new receipt. Hence, when I compared the Linked Sample to the NCCI Data, I considered only new receipt.

Of the three states that do not mandate WC coverage (South Dakota, Texas, and Wyoming), I dropped two from my analyses of the ASEC: South Dakota and Wyoming. I included Texas because [Dillender \(2015\)](#) makes a compelling case that Texas is useful to study in WC contexts. This is because Texas reports that approximately 80 percent of private-sector employees in the state are covered by WC. Conversely, there is no information available from South Dakota or Wyoming on the proportion of employees covered by WC in their respective states.

3.3 Survey of Occupational Injuries and Illnesses

The Survey of Occupational Injuries and Illnesses (SOII) is either administered by state-level agencies on behalf of the BLS or BLS staff in non-participating states. The impetus behind the survey was the signing of the Occupational Safety and Health Act of 1970. Among other provisions, the Act mandated the collection of data on OIIs. The SOII program follows record-keeping guidelines set forth by the Occupational Safety and Health Administration and includes additional information on OIIs that involved at least one day away from work (hereafter, lost-time injuries). As it relates to this paper, recipients of workers' compensation indemnity benefits should approximate a subset of lost-time cases because indemnity claims typically start when an OII necessitates a worker being away from work ([Welch et al. 2024](#)).

The SOII surveys approximately 230,000 establishments and provides them with advance notification of their *obligation* to provide data on OIIs for a specified survey year. Importantly, the Occupational Safety and Health Administration requires private firms to fill out the SOII if they receive such a notice from the BLS. This applies to firms in nonhazardous industries, which may be otherwise exempt from keeping OII records. Establishments report basic information on themselves (*e.g.*, employee hours worked), summary information on OIIs, and more detailed information on lost-time cases. Because the BLS explicitly instructs employers to copy data from their OSHA

Form 300 logs to complete the survey, the SOII serves as a reflection of regulatory reporting rather than a completely independent source of data on OIIs ([Ruser 2008](#)). Each year, the data collection process is completed by the middle of summer.

There are limitations to using the SOII data. For instance, BLS does not publish estimates for each state each year. This is because states need to participate in survey administration for state-level estimates to be released. Additionally, the BLS is aware of issues relating to undercounts of OIIs. These limitations, which are addressed in detail by [National Academies of Sciences, Engineering, and Medicine \(2018\)](#), make the SOII a potentially problematic dataset with which to work. Despite its well-known limitations, there is no other source of OII data with the scope and history of the SOII. This makes the SOII the best source of OII data for this paper.

4 How has the receipt of workers' compensation changed over time?

In the ASEC, I defined a recipient of WC to be a respondent who reported receiving a positive amount of income from WC over the past year. (This is based on the IPUMS CPS variable *INCWKCOM*.) My working assumption was that respondents were reporting indemnity benefits and not medical benefits. This seemed reasonable because indemnity benefits are paid to the recipient whereas medical benefits are generally paid to the healthcare provider directly from the WC insurer.

I estimated WC receipt as a rate and maintained internal consistency within datasets. The NCCI Data included information only on its estimate of workers insured under WC; this meant the rate for the NCCI Data is in terms of 100 insured workers. Unfortunately, the NCCI Data did not contain information on hours worked. For the ASEC and the SOII, I calculated the rates in terms of 100 full-time equivalent (FTE) employees. Because there is no information in either the ASEC or the SOII on the insurance status of workers, the rates calculated in terms of 100 FTE employees do not

consider insurance status. To calculate these rates, I used the following equation:

$$WC \text{ per } 100 \text{ FTE} = 100 \times \frac{WC}{FTE} = 100 \times \frac{WC}{H_{ASEC}} \times 2,000 \quad (1)$$

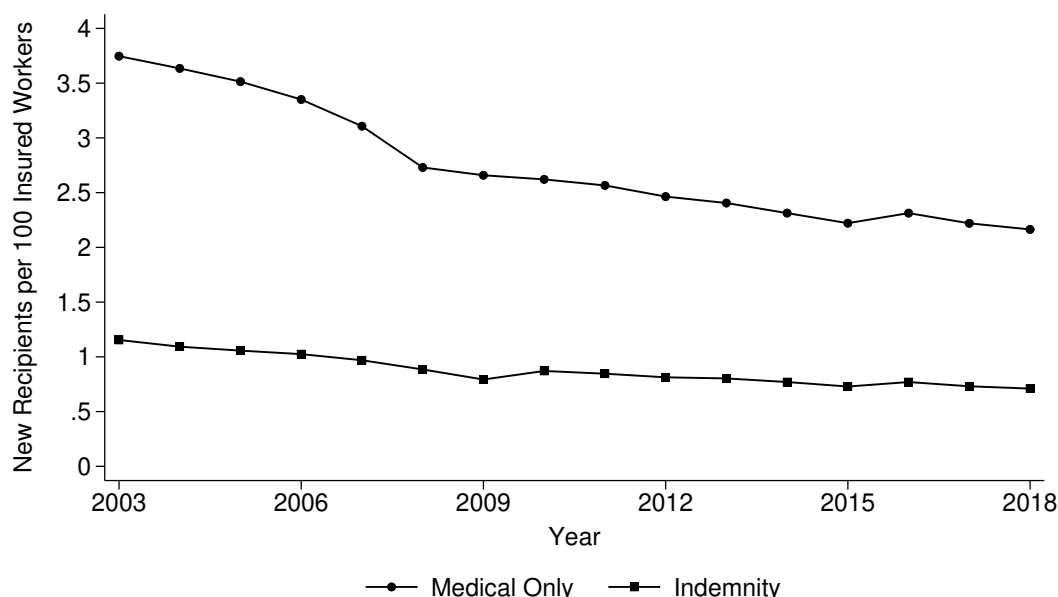
In equation 1, WC refers to the number of WC recipients, FTE refers to the number of full-time equivalent employees and H_{ASEC} refers to the total number of employee hours each year. There is no WC coverage indication in the ASEC. Therefore, I used the number of employee hours worked to proxy for the number of covered workers. This means that the denominator includes uninsured employees, potentially underestimating the rate of receipt per insured worker compared to administrative data. I obtained H_{ASEC} based on the hours usually worked per week at a respondent's main job; I chose main job because this was how I inferred the industry sector of an OII. (This is based on the IPUMS CPS variable *UHRSWRK1*.) Following BLS convention, I considered one year of work to be 50 weeks and 1 FTE employee to be an employee who works 2,000 hours. Thus, $H_{ASEC} = 50 \times \text{Hours Usually Worked per Week}$ and $FTE = H_{ASEC}/2,000$. Finally, I scaled up to 100 FTE for ease of exposition. To calculate changes over time, I subtracted the mean of the first three years of the period (2003-2005) from the mean of the last three years of the period (2016-2018). While I used this method due to the relatively few observations of WC receipt over my study period in the ASEC (mean unweighted $N = 98$), I followed this method with the other datasets in this paper, as well.

4.1 Trends by benefit type

In general, all recipients of WC benefits receive coverage of medical expenses. About 75 percent of WC recipients receive no more than this; they are called recipients of medical-only benefits. The other 25 percent of recipients additionally receive indemnity benefits; recipients receiving indemnity benefits account for about 90 percent of total benefit spending in WC programs (Welch et al. 2024).

To begin my descriptive analysis of trends, I first used the NCCI Data to examine trends in new

Figure 1: Trends in the Rates of Medical-Only and Indemnity Benefit Receipt, NCCI Data, 2003-2018

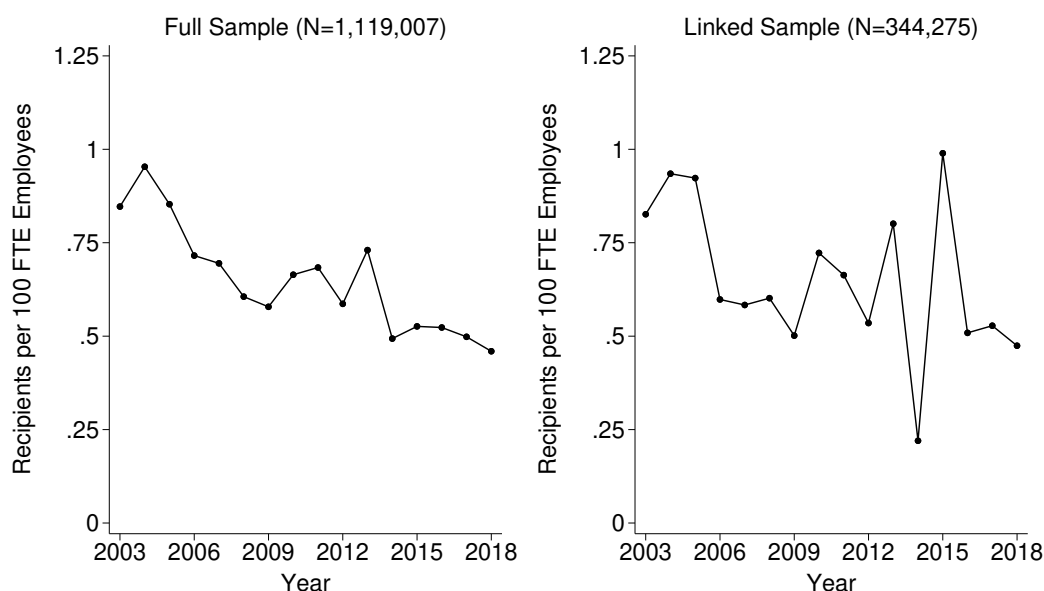


Data from the *Annual Statistical Bulletin* of the National Council on Compensation Insurance.

receipt of medical-only benefits and new receipt of indemnity benefits over the period from 2003 through 2018. Figure 1 shows these trends as the number of new recipients per 100 insured workers. The trends did not appear to exhibit any substantial degree of cyclicity at the annual level of the data. Over the period, the rate of new receipt of medical-only benefits decreased from 3.632 per 100 insured workers to 2.232 per 100 workers, a decrease of 38.5 percent. Similarly, the rate of new receipt of indemnity benefits decreased from 1.101 per 100 workers to 0.737 per 100 workers, a decrease of 33.1 percent. Hence, since the rate of new receipt of medical-only benefits decreased more than that of indemnity benefits, the proportion of all recipients who received medical-only benefits decreased slightly, from 76.5 percent to 75.3 percent, a (relative) decrease of 1.5 percent.

These results were indicative of three potential forces. First, the rate of OIIs may have decreased substantially overall, leading to the substantial observed decreased in WC receipt overall. Second, since the rate of medical-only receipt decreased more substantially than that of indemnity benefit receipt, this implied that less serious OIIs may be decreasing at a faster rate than more serious OIIs.

Figure 2: Trends in the Rate of Workers' Compensation Reciprocity, Indemnity Benefits, 2003-2018



A recipient is defined as a respondent-year with a positive amount of workers' compensation income. The Full Sample included all respondents each year, even those that cannot be linked between two years of the ASEC. The Linked Sample included the second observation of those respondents who can be successfully linked between two years of the ASEC.

This is because medical-only receipt suggests a recipient has not suffered an OII severe enough to be away from work long enough to satisfy the indemnity benefit waiting period of between three and seven days depending on the state. Finally, these findings may suggest an increasingly more stringent review process for providing medical benefits. A more stringent review process could lead to workers who experience less severe OIIs to either not file a WC application or not be awarded benefits.

4.2 Trends in indemnity benefits

Turning to analysis of the ASEC, Figure 2 shows the trend in the number of recipients per 100 FTE employees in both the Full Sample and the Linked Sample. In the Full Sample, the number of recipients per 100 FTE employees decreased from 0.884 to 0.493, a decrease of 44.2 percent. In the Linked Sample, the number of recipients per 100 FTE employees decreased from 0.894 to

0.504, a decrease of 43.6 percent. In Appendix Figure A.1, I show that similar decreases exist when considering the dollar amount of indemnity benefits per 100 FTE employees, adjusted for inflation. The results indicated a substantial decrease in indemnity benefit receipt over the period. Moreover, these findings suggested that estimates in the change over time from the Full Sample and Linked Sample were similar, differing by only 0.6 percentage points, or slightly over 1 percent of the mean of the two values. This supported my use of the Linked Sample in the calculations going forward.

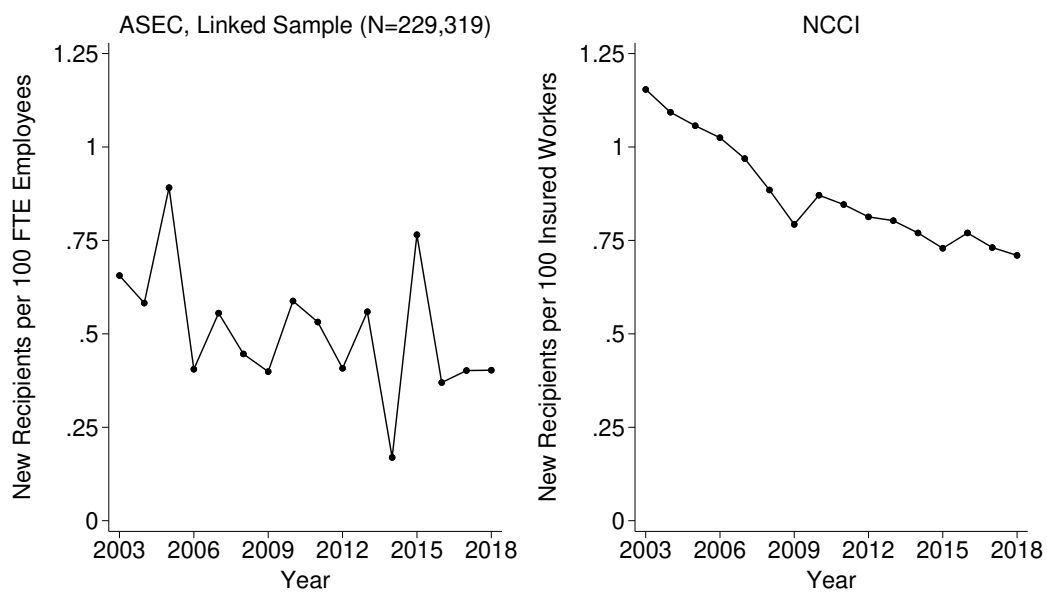
An additional comparison—between the Linked Sample and the NCCI Data—would indicate how well the ASEC captured receipt. This is because the NCCI Data is based on administrative data in the NCCI states. Whereas the analysis in the preceding paragraph considered all receipt of program benefits, the NCCI Data comprises new recipients only. Because of this, to compare the ASEC to the NCCI Data, I needed to consider only new recipients in the ASEC. This was possible only using the Linked Sample because I needed to observe a respondent switching from non-receipt to receipt to classify that respondent as a new recipient. Additionally, when comparing the Linked Sample and the NCCI Data, I restricted the Linked Sample to states that report data to the NCCI. Lastly, because of how NCCI reports these data, I included South Dakota in only this analysis.⁴

Figure 3 compares the trend of new recipients in this *modified* Linked Sample to the trend in new recipients from the NCCI Data. The rate of receipt in the modified Linked Sample decreased from 0.710 per 100 FTE employees to 0.391 per 100 FTE employees, a decrease of 44.9 percent. And the rate of receipt as calculated from the NCCI Data decreased from 1.101 per 100 insured workers to 0.737 per 100 insured workers, a decrease of 33.1 percent. Observing substantial decreases in both data sources strengthened the evidence that the decrease has actually occurred.

It is apparent from this comparison that the ASEC did not capture all receipt. This is consistent with findings that household surveys under-report WC receipt (Meyer et al. 2015). In large part, this likely stems from recall bias regarding benefits received earlier in the reference period. Furthermore, the use of proxy reporting (*i.e.*, one household member answering for another) in the ASEC may

⁴NCCI reports jurisdiction-level rates and countrywide rates per 100,000 insured workers. However, NCCI does not report the total number of insured workers by jurisdiction. This means that I cannot calculate the countrywide rate with South Dakota excluded.

Figure 3: Comparison of Trends between the ASEC and NCCI Data, New Recipients of Indemnity Benefits, 2003-2018



A recipient in the ASEC is defined as a respondent-year with a positive amount of workers' compensation income. The Linked Sample included the second observation of those respondents who can be successfully linked between two years of the ASEC. This figure further restricts the Linked Sample to states who report data to the NCCI, but includes South Dakota based on how NCCI reports these data.

lead to omissions of irregular income sources ([U.S. Census Bureau 2006](#)). Finally, respondents may misclassify WC benefits as payments from other social insurance programs due to confusion between programs ([Meyer et al. 2015](#)).

On the other hand, the NCCI Data has the advantage of accurately capturing the number of accepted applications for medical-only and indemnity benefits, for the majority of the country. But it has the disadvantage that 12 states are not captured by its data. And it does not contain industry sector-level information on where OIIs occurred, making it impossible to perform the decomposition analyses in the next section.

Overall, the analyses in this section documented a substantial decrease of 43.6 percent in total WC receipt based on estimations from the Linked Sample. In the next section, I studied the reasons that can explain why this 43.6 percent decrease occurred.

5 Why has the receipt of workers' compensation changed over time?

5.1 The role of new receipt and continued receipt

The decrease in total receipt from 0.894 per 100 FTE employees to 0.504 per 100 FTE employees was due to factors affecting both continued receipt and new receipt. Continued receipt refers to respondents to the ASEC being a recipient of WC benefits in both years. New receipt refers to respondents who are not a recipient in the first year of response, but are a recipient in the second year.

Table 1 shows the breakdown of the decrease in total receipt into both of these components. Decreases in the rate of continued receipt represented 12.1 percent of the change in total receipt. Decreases in the rate of new receipt represented 87.9 percent of the change in total receipt. These results indicated that the primary explanation for the decrease in total receipt were decreases in new receipt.

Table 1: Workers' Compensation Changes by Component, Linked Sample, 2003-2018

	(1) Rate, 2003-2005 per 100 FTE	(2) Rate, 2016-2018 per 100 FTE	(2)–(1) Rate Change per 100 FTE	Rate Change as Percent of Total Rate Change
Continued Receipt	0.116	0.069	-0.047	12.1%
New Receipt	0.777	0.435	-0.343	87.9%
Total	0.894	0.504	-0.390	100.0%

Rates are in terms of 100 FTE employees.

The rate columns are means over the sub-periods indicated, either 2003 through 2005 or 2016 through 2018.

Decomposing the decrease in total receipt into continued receipt and new receipt was essential for the analyses in the next two subsections. Specifically, analyzing new receipt allowed me to infer the industry sector of an OII based on a respondent's primary occupation in the first year of the ASEC. This inference was not possible for continued receipt because these OIIs occurred prior to the survey period. Thus, focusing on new receipt enabled me to attribute the changes in new receipt or OIIs to the specific industry sectors where the underlying OIIs occurred.

5.2 The role of sectoral composition

One plausible explanation for the decrease in new receipt was the changing sectoral composition of the economy, from presumably more hazardous goods-producing industry sectors towards safer service-providing industry sectors (Charles et al. 2019). Using new receipt, I investigated this explanation through the following decomposition:

$$\underbrace{\frac{\omega_t}{\eta_t^{ASEC}} - \frac{\omega_s}{\eta_s^{ASEC}}}_{\text{Change in Rate of New Receipt}} = \underbrace{\sum_j \left(\frac{\eta_{j,t}^{ASEC}}{\eta_t^{ASEC}} - \frac{\eta_{j,s}^{ASEC}}{\eta_s^{ASEC}} \right) \cdot \frac{\omega_{j,t}}{\eta_{j,t}^{ASEC}}}_{\text{Composition Effect}} + \underbrace{\sum_j \left(\frac{\omega_{j,t}}{\eta_{j,t}^{ASEC}} - \frac{\omega_{j,s}}{\eta_{j,s}^{ASEC}} \right) \cdot \frac{\eta_{j,s}^{ASEC}}{\eta_s^{ASEC}}}_{\text{Within-Sector Effect}} \quad (2)$$

That is, equation 2 decomposes the change in the rate of new receipt into a composition effect due to changing sectoral composition and a within-sector effect. Here, I used employment shares as a proxy for sectoral composition, and I calculated them based on the estimated hours worked as an

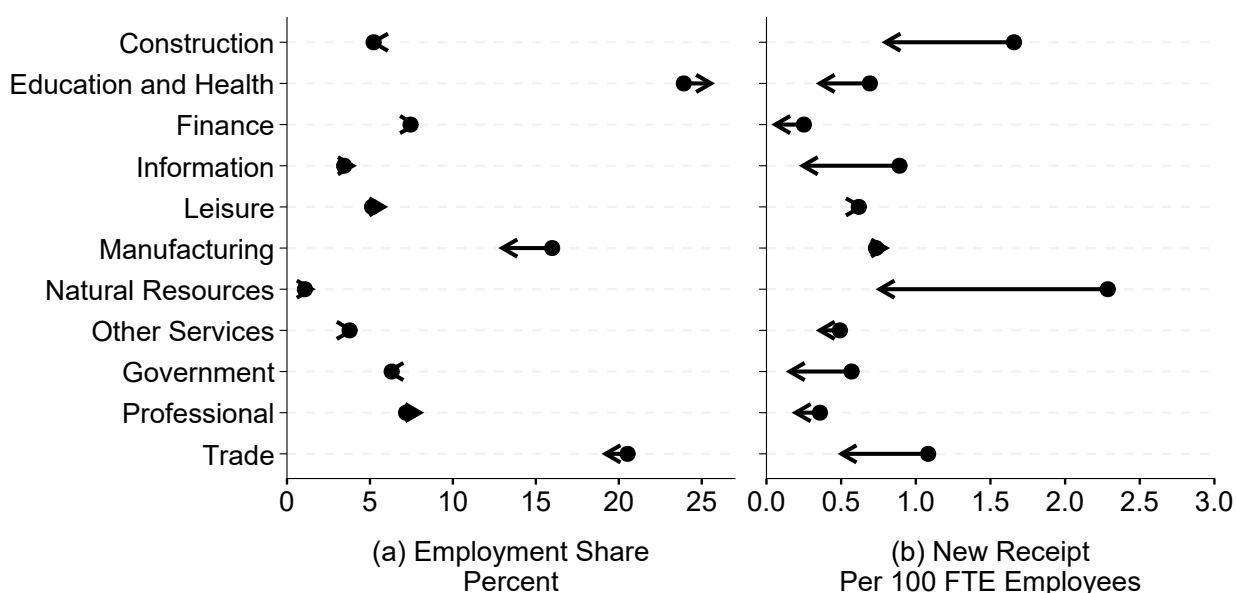
employee from the ASEC.

In equation 2, subscripts s and t refer to the following sub-periods: $s = 2003, 2004, 2005$ and $t = 2016, 2017, 2018$. Sectors are indexed by j . ω is the count of new receipt in sub-period s or t and, where indicated by subscript, for sector j . η^{ASEC} is the count of employee hours in sub-period s or t and, where indicated by subscript, for sector j . I obtained η^{ASEC} based on the hours usually worked per week at a respondent's main job; I chose main job because this was how I inferred the industry sector of an OIL. The first part of the sum on the right hand side is the effect of an absolute change in employment share weighted by the ending within-sector rate of new receipt (*composition effect*). The second part of the sum on the right hand side is the effect of a change in the within-sector rate of new receipt weighted by the beginning employment share (*within-sector effect*). The counterfactual term $\frac{\omega_{j,t}}{\eta_{j,t}} \cdot \frac{\eta_{j,s}}{\eta_s}$ that is added and subtracted on the right hand side to obtain the decomposition, represents the ending within-sector rates of new receipt weighted by the beginning sectoral composition. Summing over j would provide the ending overall rate of new receipt had the sectoral composition remained constant. The components of the decomposition are not causal estimates. However, without an exogenous shock, it is unlikely that causal estimates could be obtained in this context under any empirical approach.

I performed the decomposition based on the industry sectors that the BLS refers to as super-sectors. I refer to these supersectors as *sectors* in this paper. Table A.1 in the Appendix shows a crosswalk between the sectors in this paper and two-digit NAICS codes. To maintain consistency with the next Subsection, and due to the relatively small sample size of new receipt each year (yearly mean unweighted $N = 98$), performing the decomposition down to more detailed industry identifiers was not feasible.

Figure 4 displays the sector-level components that enter into the calculation of both the composition and within-sector effects. Appendix Table A.2 includes all of the underlying numerical values in the calculation. Sub-figure (a) shows the beginning and ending employment share values as percentages, with arrows indicating the direction of the change. Sub-figure (b) does the same for new receipt per 100 FTE employees.

Figure 4: Components of the Decomposition Calculation, Change in Rate of New Receipt of Workers' Compensation Benefits, 2003-2018



Results: *Change in Rate of New Receipt:* -0.343 per 100 FTE employees.
Composition Effect: -0.014 per 100 FTE employees. *Within-Sector Effect:* -0.329 per 100 FTE employees.

Changes in sectoral composition occurred primarily due to shifts in employment away from the Manufacturing and Trade sectors and towards the Education and Health and Professional sectors. By contrast, five sectors experienced an increase or decrease in employment share of less than 0.5 percentage points: Construction, Finance, Natural Resources, Other Services, and Government. However, these changes were not weighted equally in the calculation. Construction, Leisure, Manufacturing, Natural Resources, and Trade all had ending rates of new receipt greater than 0.5 per 100 FTE employees. Hence, the relatively small employment changes in the Construction and Natural Resources sectors received approximately the same weight as the relatively large changes in the Leisure, Manufacturing, and Trade sectors. It was not surprising, then, that the estimated composition effect was relatively small: -0.014 per 100 FTE employees, or 4.1 percent of the -0.343 per 100 FTE employees change in new receipt over the period. This result suggested a minimal role for changing sectoral composition in driving the decrease in new receipt.⁵

⁵This interpretation is robust to swapping the time periods of the weights in the calculation. These changes would

All sectors except two, Leisure and Manufacturing, experienced decreases in within-sector rates of new receipt. Notably, the increases in those two sectors were less in magnitude than the decrease in any other sector. Moreover, those sectors comprised approximately 21 percent of employment at the beginning of the period. In combination, this offset the calculation of the within-sector effect by 0.012 per 100 FTE employees. Overall, the estimated within-sector effect was -0.329 per 100 FTE employees, or 95.9 percent of the -0.343 per 100 FTE employees change in new receipt over the period. This estimate implied that the changes in new receipt resulted almost completely from within-sector decreases.

One limitation of this analysis was the relatively high level of aggregation. Consequently, my approach excluded the influence of the changing composition of occupations within these sectors. I could not analyze these data at more detailed industry classifications due to the limited number of observations of receipt in the ASEC. However, I address this issue in the next subsection, where I perform decompositions of OIIs at both the sector level and the more detailed industry level.

5.3 The role of occupational injuries and illnesses

Another plausible explanation for the decrease in new receipt was the long-term decrease in OIIs (Cohn and Wardlaw 2024). Because the ASEC reports the receipt of WC indemnity benefits and WC indemnity benefits generally require workers to be out of work initially, OIIs involving days away from work (hereafter, lost-time injuries) were the most appropriate type of OII to study.

To understand the factors associated with the decrease in new receipt, I performed a decomposition analogous to equation 2 on the rates of lost-time injuries:

$$\underbrace{\frac{\delta_t}{\eta_t^{SOII}} - \frac{\delta_s}{\eta_s^{SOII}}}_{\text{Change in Rate of Lost-Time Injuries}} = \underbrace{\sum_j \left(\frac{\eta_{j,t}^{SOII}}{\eta_t^{SOII}} - \frac{\eta_{j,s}^{SOII}}{\eta_s^{SOII}} \right) \cdot \frac{\delta_{j,t}}{\eta_{j,t}^{SOII}}}_{\text{Composition Effect}} + \underbrace{\sum_j \left(\frac{\delta_{j,t}}{\eta_{j,t}^{SOII}} - \frac{\delta_{j,s}}{\eta_{j,s}^{SOII}} \right) \cdot \frac{\eta_{j,s}^{SOII}}{\eta_s^{SOII}}}_{\text{Within-Sector Effect}} \quad (3)$$

result in a composition effect of -0.004 per 100 FTE employees and a within-sector effect of -0.339 per 100 FTE employees.

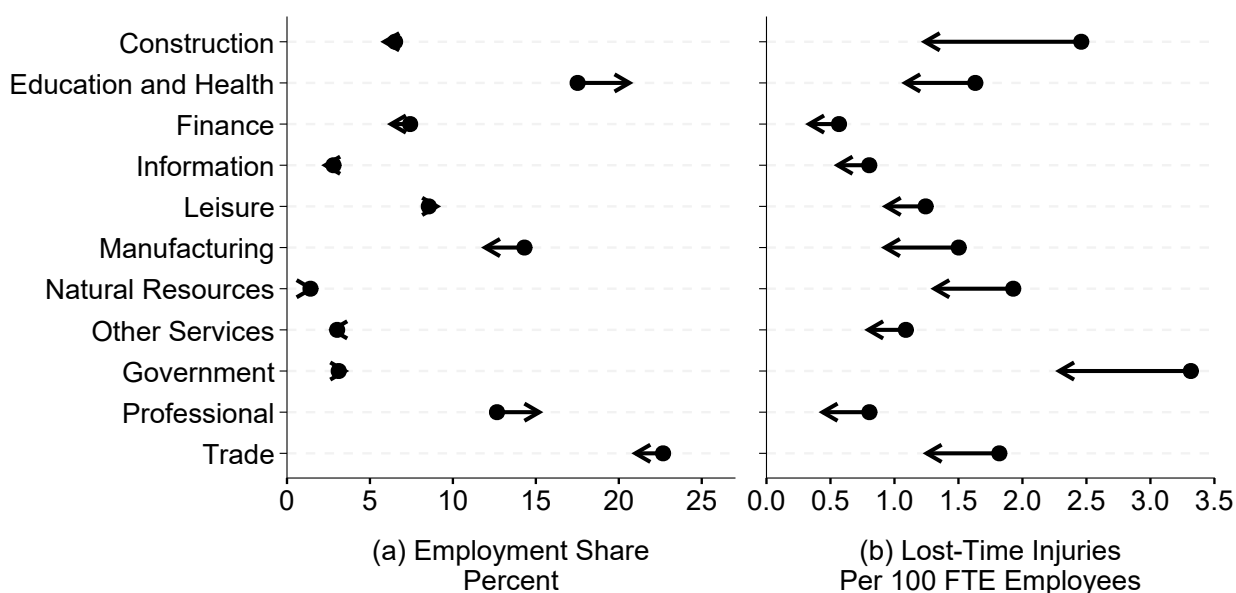
Similar to equation 2, equation 3 decomposes the change in the rate of lost-time injuries into a composition effect and a within-sector effect. Again, I used employment shares as a proxy for sectoral composition, but here I estimated hours from within the SOII to be internally consistent within the dataset.

In equation 3, subscripts s and t refer to the following sub-periods: $s = 2003, 2004, 2005$ and $t = 2016, 2017, 2018$. Sectors are indexed by j . δ is the count of lost-time injuries in sub-period s or t and, where indicated by subscript, for sector j . η^{SOII} is the count of employee hours in sub-period s or t and, where indicated by subscript, for sector j . η^{SOII} is back-calculated from the estimates of incidence rates in the SOII. The first part of the sum on the right hand side is the effect of an absolute change in employment share weighted by the ending within-sector rate of lost-time injuries (*composition effect*). The second part of the sum on the right hand side is the effect of a change in the within-sector rate of lost-time injuries weighted by the beginning employment share (*within-sector effect*). The counterfactual term $\frac{\delta_{j,t}}{\eta_{j,t}} \cdot \frac{\eta_{j,s}}{\eta_s}$, that is added and subtracted on the right hand side to obtain the decomposition, represents the ending within-sector rates of lost-time injuries weighted by the beginning sectoral composition. Summing over j would provide the ending overall rate of lost-time injuries had the sectoral composition remained constant. The components of the decomposition are not causal estimates. However, without an exogenous shock, it is unlikely that causal estimates could be obtained in this context under any empirical approach.

Figure 5 displays the sector-level components that enter into the calculation of both the composition and within-sector effects. Appendix Table A.3 includes all of the underlying numerical values in the calculation. Sub-figure (a) shows the beginning and ending employment share values as percentages, with arrows indicating the direction of the change. Sub-figure (b) does the same for lost-time injuries per 100 FTE employees.

Changes in sectoral composition occurred primarily due to shifts in employment away from the Finance, Manufacturing and Trade sectors and towards the Education and Health and Professional sectors. Interestingly, the Finance sector exhibited a substantial decrease in employment share when estimated from the SOII data (from 7.4 percent to 6.2 percent), yet a relatively small increase

Figure 5: Components of the Decomposition Calculation, Change in Rate of Lost-Time Injuries, 2003-2018

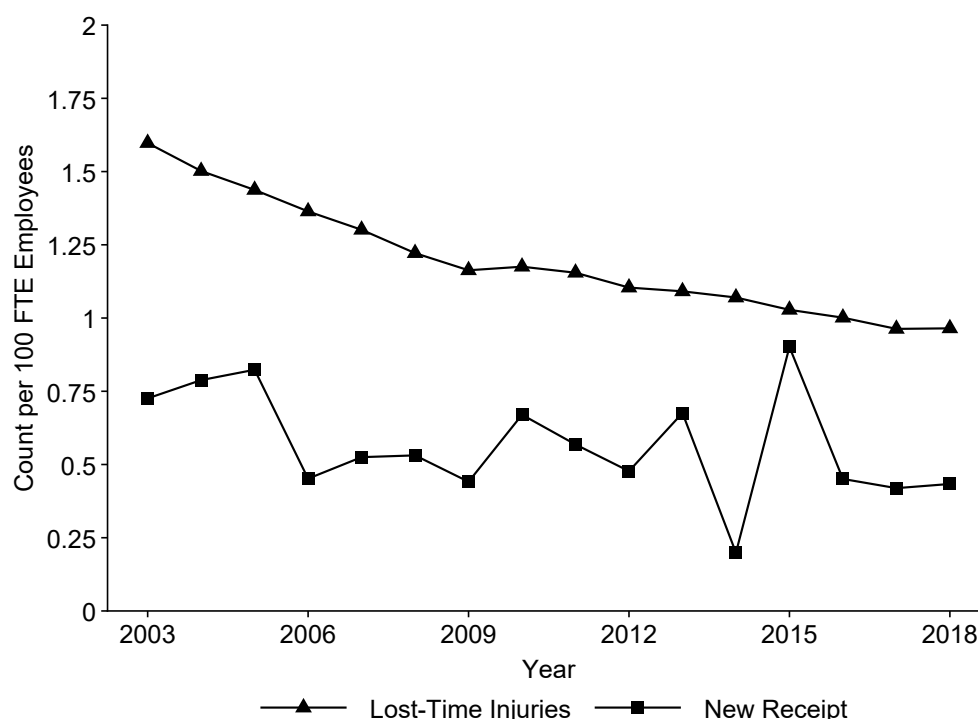


Results: *Change in Rate of Lost-Time Injuries:* -0.534 per 100 FTE employees.
Composition Effect: -0.000 per 100 FTE employees. *Within-Sector Effect:* -0.533 per 100 FTE employees.

in employment share when estimated from the ASEC data (from 7.4 percent to 7.7 percent). This occurred despite the sector starting at similar shares of employment in the two datasets. Along these lines, five sectors experienced an increase or decrease in employment share of less than 0.5 percentage points, but these also did not perfectly coincide with the estimates from the ASEC: Information, Leisure, Natural Resources, Other Services, and Government. Such discrepancies likely resulted from the differing survey methodologies of each sample—a voluntary household-based survey (ASEC) versus a mandatory firm-based survey (SOII). The estimated composition effect from the SOII was tiny: -0.000 per 100 FTE employees, or merely 0.1 percent of the -0.534 per 100 FTE employees change in lost-time injuries over the period. This means there was an almost nonexistent role for changing sectoral composition driving the decrease in the overall rate of lost-time injuries.⁶

⁶This interpretation is robust to swapping the time periods of the weights in the calculation. These changes would result in a composition effect of -0.004 per 100 FTE employees and a within-sector effect of -0.530 per 100 FTE employees.

Figure 6: Trends in the Rates of Lost-Time Injuries and New Workers' Compensation Receipt, 2003-2018



Every sector experienced substantial decreases in the rate of lost-time injuries, with the largest decreases seen in the Construction and Government sectors. The largest weights in calculating the within-sector effect, however, were in the Education and Health and Trade sectors. Based on these factors, the estimated within-sector effect from the SOII was -0.533 per 100 FTE employees, which explains over 99.9 percent of the -0.534 per 100 FTE employees change in lost-time injuries over the period. From this estimate, I concluded that almost all of the change in the rate of lost-time injuries was a result of within-sector effects.

To assess whether the level of aggregation influenced my results, I repeated the decomposition using 85 three-digit NAICS codes across the 11 sectors. The counts of both lost-time injuries and FTE employees in this analysis were slightly lower than in the sector-level analysis, likely due to data suppression intended to protect confidentiality at more detailed industry levels. Consequently, I estimated a change in lost-time injuries of -0.548 per 100 FTE employees (versus -0.534 per 100 FTE employees at the sector level). When calculated using these values, the results showed an

even stronger within-sector effect. Specifically, I estimated a composition effect of 0.034 per 100 FTE employees. This positive value implies that shifts in employment composition slightly offset the decrease in the rate of lost-time injuries. Thus, the estimated within-sector effect of -0.582 per 100 FTE employees was associated with more than 100 percent of the total decrease in the rate of lost-time injuries.

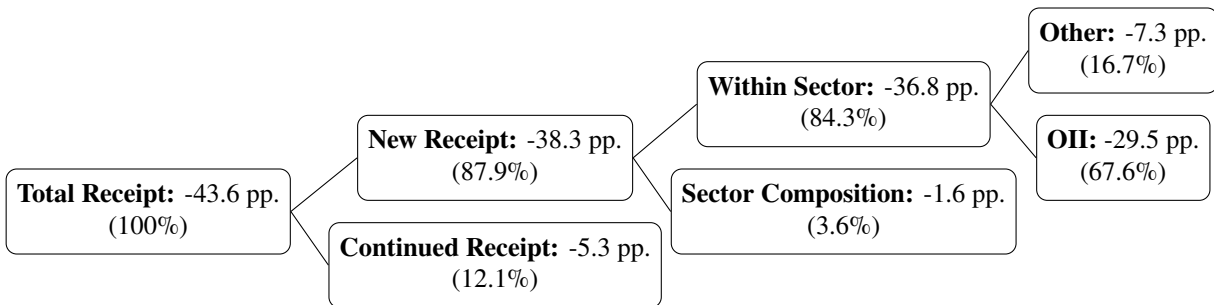
Having established that the decrease in OIIs was overwhelmingly within-sector, I then compared the trend in the rate of lost-time injuries to the trend in the rate of new WC receipt. This comparison allowed me to determine how much of the decrease in the rate of new receipt could be explained by the decrease in the rate of lost-time injuries. To maintain internal consistency within the SOII dataset, I calculated the rate of lost-time injuries using hours estimated directly from the SOII data. Analogous to Equation 1, the rate is calculated as the following:

$$\text{Lost-Time Injuries per 100 FTE} = 100 \times \frac{\text{Injuries}}{H_{SOII}} \times 2,000 \quad (4)$$

In this equation, *Injuries* refers to the count of OIIs involving days away from work, and H_{SOII} represents the total employee hours back-calculated from the SOII estimates of incidence rates.

Figure 6 shows the trend in the number of lost-time injuries per 100 FTE employees and the trend in the number of new WC receipt per 100 FTE employees over the study period of 2003 through 2018. Over the period, the rate of lost-time injuries decreased from 1.510 to 0.976, a decrease of 35.3 percent. The rate of new WC receipt decreased from 0.777 to 0.435, a decrease of 44.1 percent. Taken together, this implied that 80.2 percent ($= 35.3/44.1 \times 100\%$) of the decrease in new WC receipt could be explained by the decrease in the rate of lost-time injuries. This association could have occurred due to genuine improvements to workplace safety from sources such as increased use of automation and new regulations from the Occupational Safety and Health Administration. Alternatively, the association could reflect employers becoming more aggressive at returning employees to work or minimizing recordable OIIs.

Figure 7: Breakdown of Relative Percent Change in Workers' Compensation Receipt



pp. are contributions in percentage points of the specified component to the total relative percent change. Percents in parentheses show percent of total relative percent change. Sums may not equal totals due to rounding.

OII: Occupational injury and illness.

5.4 Synthesis of results

Figure 7 brings together the results from this section via a breakdown of the 43.6 percentage point decrease in the rate of total WC receipt from 2003 through 2018. The figure shows that the most important contributor to the decrease was the broad-based decrease in lost-time injuries, which contributed 29.5 percentage points, or 67.6 percent, of the total. However, the analysis cannot separate genuine improvements in workplace safety from changes in reporting behavior. Hence, this component reflects the net impact of improvements to workplace safety and more aggressive claims management. Since I determined that this decrease in OIIs occurred almost entirely within-sector, it is informative to consider possible reasons why these decreases occurred. This is especially true in light of the descriptive nature of my analyses that does not show the causal impact of any factor directly.

First, decreases in OIIs may have occurred due to technological improvements. [Cohn and Ward-law \(2024\)](#) discuss the increased adoption of automation, robotics, and smart sensors that reduce the hazards that workers encounter. This is especially true in the Manufacturing and Construction sectors. Additionally, decreases may have occurred due to implementation of regulations from the federal Occupational Safety and Health Administration (OSHA).⁷ The regulations adopted over

⁷While there are 22 states that administer their own occupational safety and health programs, federal law requires that these state-level plans must be *at least as effective* as federal OSHA.

the period by federal OSHA include requiring employers to pay for personal protective equipment as of 2008, establishing a Severe Violator Enforcement Program for repeat offenders in 2010, and mandating uniform safety data sheets for hazardous chemicals as of 2015.

Employer behavior and reporting incentives likely contribute to the decrease in OII rates. Financial incentives to minimize WC claims (to reduce premiums) often align with regulatory incentives to minimize incidents reported on OSHA logs (to avoid inspections) ([Azaroff et al. 2002](#); [Rappin et al. 2016](#)). This alignment is critical because the BLS instructs employers to copy information directly from their OSHA logs to the SOII ([Ruser 2008](#)). While there is no 1:1 link between a WC claim and an SOII response ([Boden and Ozonoff 2008](#)), employers increasingly utilize third-party consultants and aggressive claims management strategies that result in the reduction of both. For instance, aggressive return-to-work programs are designed to disqualify workers from indemnity benefits by bringing them back to light-duty work immediately ([Pransky et al. 1999](#)). Because a case only becomes a lost-time injury on OSHA logs if the worker misses at least one full day of work, such programs can simultaneously eliminate a WC indemnity claim and prevent the incident from being classified as a lost-time injury in the SOII data ([Boden and Ruser 2003](#)). Therefore, the OII component in my decomposition results (29.5 percentage points) should be interpreted as capturing the net impact of both genuine safety improvements and these behavioral reporting factors, as my analysis cannot disentangle the two.

The least important contributor to the change in total receipt was changing sectoral composition, which can explain only 1.6 percentage points, or 3.6 percent, of the total decrease. This result is notable given the structural transformation of the economy over this period. Specifically, employment shares shifted away from hazardous goods-producing sectors, such as Manufacturing, and toward relatively safer service-providing sectors, such as Education and Health Services. While this shift did exert downward pressure on receipt, it was overshadowed by the substantial decreases in injury rates that occurred within almost every sector. As discussed in Subsection 5.2, this finding is consistent with results in the literature that rely on more detailed occupational data ([Cohn and Wardlaw 2024](#)).

Changes in continued receipt contributed 5.3 percentage points, or 12.1 percent. Factors affecting continued receipt imply that the duration for which workers are receiving benefits has decreased over time. These decreases could have occurred for a variety of reasons. One possible explanation was that the severity of OIIs decreased over the period. That is, as OIIs became less severe, workers received WC benefits for a shorter duration. Another possible explanation was a decrease in the generosity of the program for the characteristics of a given recipient, indicating that the program was providing benefits for a shorter duration.

Lastly, what could be the “other” factors, accounting for 7.3 percentage points, or 16.7 percent, of the decrease in total receipt? Prior work has found that fear of retaliation for applying for WC benefits may decrease receipt. Approximately 20 percent of surveyed workers who lost more than seven days of work due to an OII responded that they did not apply for WC benefits due to fear of retribution ([Biddle et al. 1998](#)). Additionally, union workers, who have stronger job protections than nonunion workers, have been found 60 percent more likely to receive WC benefits than similar nonunion workers ([Hirsch et al. 1997](#)). Given the well-known decline in unionization over time ([Pew Research Center 2025](#)), an increase in application hesitancy over the study period could plausibly have served as a factor that reduces receipt. Increased access to other safety net programs may have contributed to the decrease in receipt of WC, as well. For instance, [Dong et al. \(2024\)](#) estimated through a difference-in-differences approach that implementation of mandated paid sick leave reduced the likelihood of benefit receipt by 13.5 percent. Since 11 states implemented paid sick leave during the study period, this could also plausibly have reduced receipt. Continuing to explore the impact of factors like these can serve as the basis for future research.

6 Conclusion

In this paper, I systematically investigated the reasons for the long-term decrease in WC receipt over the period from 2003 through 2018. My analysis of the NCCI Data showed that the rate of new receipt for medical-only benefits decreased by 38.5 percent and for indemnity benefits by 33.1

percent. The more substantial decrease in medical-only receipt may imply that less serious OIIs were decreasing at a faster rate than more serious ones. To understand these trends in more depth, I turned to the ASEC, which provides data on all states and allows for a study at the sector-level. I showed that the rate of receipt decreased by 43.6 percent over the study period. Then, I used decomposition analyses to first attribute the vast majority of this decrease—87.9 percent—to a decrease in the rate of new receipt. This was driven almost entirely by within-sector changes rather than shifts in sectoral composition. The primary mechanism for this change was a decrease in lost-time injuries. This ultimately explained 67.6 percent of the total decrease in receipt. My results suggested the decrease in WC receipt was primarily associated with improvements to workplace safety.

References

- Abouk, R., Ghimire, K. M., Maclean, J. C., and Powell, D. (2023). Pain Management and Work Capacity: Evidence From Workers' Compensation and Marijuana Legalization. *Journal of Policy Analysis and Management*, 42(3):737–770.
- Asfaw, A., Pana-Cryan, R., and Rosa, R. (2011). The business cycle and the incidence of workplace injuries: Evidence from the U.S.A. *Journal of Safety Research*, 42(1):1–8.
- Azaroff, L. S., Levenstein, C., and Wegman, D. H. (2002). Occupational injury and illness surveillance: Conceptual filters explain underreporting. *American Journal of Public Health*, 92(9):1421–1429.
- Biddle, J. (2001). Do High Claim-Denial Rates Discourage Claiming? Evidence from Workers Compensation Insurance. *Journal of Risk and Insurance*, 68(4):631–658.
- Biddle, J. and Roberts, K. (2003). Claiming Behavior in Workers' Compensation. *Journal of Risk and Insurance*, 70(4):759–780.
- Biddle, J., Roberts, K., Rosenman, K., and Welch, E. (1998). What Percentage of Workers With Work-Related Illnesses Receive Workers' Compensation Benefits? *Journal of Occupational and Environmental Medicine*, 40(4):325–330.
- Boden, L. I. and Ozonoff, A. (2008). Capture-recapture estimates of nonfatal workplace injuries and illnesses. *Annals of Epidemiology*, 18(6):500–506.
- Boden, L. I. and Ruser, J. W. (2003). Workers' Compensation “Reforms”, Choice of Medical Care Provider, and Reported Workplace Injuries. *The Review of Economics and Statistics*, 85(4):923–929.

- Bronchetti, E. T. and McInerney, M. (2012). Revisiting Incentive Effects in Workers' Compensation: Do Higher Benefits Really Induce More Claims? *ILR Review*, 65(2):286–315.
- Charles, K. K., Hurst, E., and Schwartz, M. (2019). The Transformation of Manufacturing and the Decline in US Employment. *NBER Macroeconomics Annual*, 33:307–372.
- Charles, K. K., Johnson, M. S., Stephens, M., and Lee, D. Q. (2022). Demand Conditions and Worker Safety: Evidence from Price Shocks in Mining. *Journal of Labor Economics*, 40(1):47–94.
- Cohn, J. B. and Wardlaw, M. (2024). The Decline of U.S. Workplace Injuries, Improvements in Safety, and Changes in Occupational Composition. *SSRN Electronic Journal*.
- Dillender, M. (2015). The effect of health insurance on workers' compensation filing: Evidence from the affordable care act's age-based threshold for dependent coverage. *Journal of Health Economics*, 43:204–228.
- Dong, X. and Jestrab, R. (2022). Unemployment and workplace injuries: evidence from California. *Applied Economics Letters*, 29(21):2040–2043.
- Dong, X., Maclean, J. C., and Powell, D. (2024). Social Insurance Spillovers: Evidence From Paid Sick Leave Mandates and Workers' Compensation. *NBER Working Paper*.
- Fishback, P. V. and Kantor, S. E. (2000). *A Prelude to the Welfare State: The Origins of Workers' Compensation*. University of Chicago Press.
- Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R., Backman, D., Chen, A., Cooper, G., Richards, S., Schouweiler, M., and Westberry, M. (2024). IPUMS CPS: Version 12.0 [dataset].
- Ghimire, K. M. and Maclean, J. C. (2020). Medical marijuana and workers' compensation claiming. *Health Economics*, 29(4):419–434.
- Hartwig, R. P., Retterath, R. C., Restrepo, T. E., and Kahley, W. J. (1997). Workers' Compensation and Economic Cycles: A Longitudinal Approach. *Proceedings of the Casualty Actuarial Society*.
- Hirsch, B. T., Macpherson, D. A., and DuMond, J. M. (1997). Workers' Compensation Reciprocity in Union and Nonunion Workplaces. *Industrial and Labor Relations Review*, 50(2):213–236.
- Meyer, B. D., Mok, W. K., and Sullivan, J. X. (2015). Household surveys in crisis. *Journal of Economic Perspectives*, 29(4):199–226.
- National Academies of Sciences, Engineering, and Medicine (2018). *A Smarter National Surveillance System for Occupational Safety and Health in the 21st Century*. National Academies Press, Washington, D.C.
- Pew Research Center (2025). Majorities of adults see decline of union membership as bad for the U.S. and working people.
- Pransky, G., Snyder, T., Dembe, A., and Himmelstein, J. (1999). Under-reporting of work-related disorders in the workplace: a case study and review of the literature. *Ergonomics*, 42(1):171–182.

- Rappin, C. L., Wuellner, S. E., and Bonauto, D. K. (2016). Employer reasons for failing to report eligible workers' compensation claims in the BLS survey of occupational injuries and illnesses. *The Review of Economics and Statistics*, 59(5):343–356.
- Ruser, J. W. (2008). Examining evidence on whether BLS undercounts workplace injuries and illnesses. *Monthly Labor Review*, 131:20–32.
- Ruser, J. W. (2014). Industry contributions to aggregate workplace injury and illness rate trends: 1992–2008. *American Journal of Industrial Medicine*, 57(10):1149–1164.
- U.S. Census Bureau (2006). *Current Population Survey: Design and Methodology (Technical Paper 66)*. U.S. Government Printing Office, Washington, DC. Chapter 7.
- Welch, T. Q., Murphy, G. T., and Manley, M. (2024). Workers' Compensation: Benefits, Costs, and Coverage. *National Academy of Social Insurance*.

Appendix

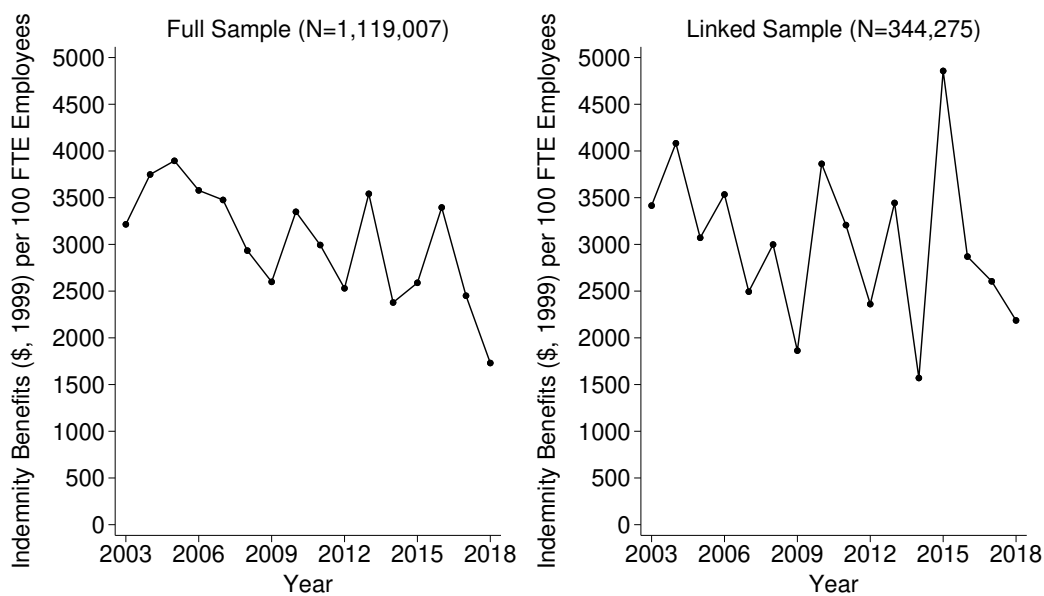
Figure [A.1](#) shows the amount of indemnity benefits, adjusted for inflation (base year = 1999), over the period from 2003 through 2018. In the Full Sample, the amount of indemnity benefits decreased from \$3,563 per 100 FTE employees to \$2,767 per 100 FTE employees, a decrease of 22.3 percent. In the Linked Sample, the amount of indemnity benefits decreased from \$3,698 per 100 FTE employees to \$2,843 per 100 FTE employees, a decrease of 23.1 percent.

Table [A.1](#) shows a crosswalk between sectors in the paper and two-digit NAICS codes.

Table [A.2](#) shows the numerical values in the calculation of the composition and within-sector effects of the change in the rate of new receipt.

Table [A.3](#) shows the numerical values in the calculation of the composition and within-sector effects of the change in the rate of DAFW OII.

Figure A.1: Trends in the Amount of, Indemnity Benefits, 2003-2018



A recipient is defined as a respondent-year with a positive amount of workers' compensation income. The Full Sample includes all respondents each year, even those that cannot be linked between two years of the ASEC. The Linked Sample includes the second observation of those respondents who can be successfully linked between two years of the ASEC.

Table A.1: Sector Definitions

Sector	NAICS Two-Digit Sector
<i>Construction</i>	Construction: NAICS 23
<i>Education and Health</i>	Educational Services: NAICS 61 Health Care and Social Assistance: NAICS 62
<i>Finance</i>	Finance and Insurance: NAICS 52 Real Estate and Rental and Leasing: NAICS 53
<i>Information</i>	Information: NAICS 51
<i>Leisure</i>	Arts, Entertainment, and Recreation: NAICS 71 Accommodation and Food Services: NAICS 72
<i>Manufacturing</i>	Manufacturing: NAICS 31-33
<i>Natural Resources</i>	Agriculture, Forestry, Fishing and Hunting: NAICS 11 Mining, Quarrying, and Oil and Gas Extraction: NAICS 21
<i>Other Services</i>	Other Services (except Public Administration): NAICS 81
<i>Government</i>	Government
<i>Professional</i>	Professional, Scientific, and Technical Services: NAICS 54 Management of Companies and Enterprises: NAICS 55 Administrative and Support and Waste Management and Remediation Services: NAICS 56
<i>Trade</i>	Wholesale Trade: NAICS 42 Retail Trade: NAICS 44-45 Transportation and Warehousing: NAICS 48-49 Utilities: NAICS 22

Sector definitions based on Supersector classifications by the Bureau of Labor Statistics.

Table A.2: Results of Decomposition of Change in Rate of New Receipt, 2003-2018

<i>Total Change in Rate of New Receipt = -0.343 per 100 FTE employees</i>			
Panel A: Composition Effect			
Sector	(1) Absolute Change in Emp. Share %	(2) Ending New Receipt per 100 FTE	(1)×(2) Composition Effect Contribution per 100 FTE
Construction	-0.031	0.798	0.000
Education and Health	1.643	0.357	0.006
Finance	0.267	0.059	0.000
Information	0.525	0.244	0.001
Leisure	0.728	0.632	0.005
Manufacturing	-3.000	0.801	-0.024
Natural Resources	0.411	0.758	0.003
Other Services	0.124	0.357	0.000
Government	-0.191	0.156	0.000
Professional	0.848	0.196	0.002
Trade	-1.325	0.503	-0.007
<i>Composition Effect</i>			-0.014
<i>Composition Effect as Percent of Total Change</i>			4.1%
Panel B: Within-Sector Effect			
Sector	(1) Absolute Change in New Receipt per 100 FTE	(2) Beginning Emp. Share %	(1)×(2) Within-Sector Effect Contribution per 100 FTE
Construction	-0.860	5.220	-0.045
Education and Health	-0.336	23.916	-0.080
Finance	-0.191	7.447	-0.014
Information	-0.647	3.443	-0.022
Leisure	0.013	5.126	0.001
Manufacturing	0.067	15.984	0.011
Natural Resources	-1.528	1.078	-0.016
Other Services	-0.135	3.773	-0.005
Government	-0.414	6.309	-0.026
Professional	-0.162	7.175	-0.012
Trade	-0.580	20.528	-0.119
<i>Within-Sector Effect</i>			-0.329
<i>Within-Sector Effect as Percent of Total Change</i>			95.9%

Table A.3: Results of Decomposition of Change in Rate of Days Away from Work Occupational Injuries and Illnesses, 2003-2018

<i>Total Change in Rate of DAFW OIIs = -0.534 per 100 FTE employees</i>				
Panel A: Composition Effect				
Sector	(1) Absolute Change in Emp. Share %	(2) Ending DAFW OIIs per 100 FTE	(1)×(2) Composition Effect Contribution per 100 FTE	
Construction	-0.646	1.237	-0.008	
Education and Health	3.122	1.085	0.034	
Finance	-1.152	0.334	-0.004	
Information	-0.497	0.556	-0.003	
Leisure	0.498	0.932	0.005	
Manufacturing	-2.372	0.926	-0.022	
Natural Resources	0.063	1.312	0.001	
Other Services	-0.290	0.795	-0.002	
Government	0.389	2.287	0.009	
Professional	2.551	0.439	0.011	
Trade	-1.666	1.252	-0.021	
<i>Composition Effect</i>			0.000	
<i>Composition Effect as Percent of Total Change</i>			0.1%	
Panel B: Within-Sector Effect				
Sector	(1) Absolute Change in DAFW OIIs per 100 FTE	(2) Beginning Emp. Share %	(1)×(2) Within-Sector Effect Contribution per 100 FTE	
Construction	-1.222	6.516	-0.080	
Education and Health	-0.547	17.518	-0.096	
Finance	-0.233	7.423	-0.017	
Information	-0.247	2.802	-0.007	
Leisure	-0.311	8.548	-0.027	
Manufacturing	-0.576	14.313	-0.082	
Natural Resources	-0.616	1.415	-0.009	
Other Services	-0.294	3.022	-0.009	
Government	-1.029	3.117	-0.032	
Professional	-0.366	12.658	-0.046	
Trade	-0.567	22.668	-0.129	
<i>Within-Sector Effect</i>			-0.533	
<i>Within-Sector Effect as Percent of Total Change</i>			99.9%	